

Correctness review: issues and minor points

Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with Log-Ratio Variation

August 27, 2026

Summary

Every displayed identity was re-derived and every number checkable from the text was recomputed; the full list is in Appendix A. **The mathematics is sound.** All four proofs — the Fisher–Rao gradient of $\mathcal{X}_\omega$, the dissipation identity, the accuracy theorem and the propagation theorem — go through as written, and the reported numbers are internally consistent to an unusual degree. Nothing below invalidates a result. What follows is a set of qualifications that ought to be stated, one dangling cross-reference, and small inconsistencies.

#	location	item	kind
1	accuracy appendix / §2.3	dangling reference; the one-nat cap is unstated	substantive
2	abstract, contribution 4	unbiasedness stated without its hypothesis	substantive
3	domination remark, FR flow	sufficient condition excludes the defaults	substantive
4	abstract, §2.2	the e^{-t} rate is forward KL’s, not KLXX’s	presentation
5	§6.5	finite-sample story applied to one target only	consistency
6	§6.1	q used for two different things	notation
7	propagation theorem, §7	theorem attached to an unnumbered scheme	precision
8	§4.3, staging algorithm	the stated default τ_v is never used	consistency

Reference key

To avoid guessing numbers, manuscript objects are named below and mapped here to their labels.

name used in this note	label in the manuscript
the accuracy theorem	thm: accuracy
the propagation theorem	thm: propagation
the variation lemma	lem: appendix-variation
the correlation lemma	lem: appendix-correlation
the one-step lemma	lem: onestep
the domination remark	rem: appendix-domination
the FR flow	eq: FR-flow
the dissipation identity	eq: combined-rate
the exponential decay	eq: KL-exponential
the stationarity identity	eq: surrogate-stationary
the accuracy bound	eq: accuracy-bound
the surrogate identity	eq: appendix-rho-minus-one
the propagation bound	eq: propagation-bound
the Amber energy	eq: molecular-amber-energy
the reduced potential	eq: molecular-reduced-potential
the staging algorithm	alg: boltzmann
the accuracy appendix	sec: appendix-accuracy
the propagation appendix	sec: appendix-propagation
the factor appendix	subsec: appendix-factor

1 Issues, by importance

1.1 A dangling reference, and a cap on the accuracy theorem that should be stated

Where. The accuracy appendix, paragraph after the proof; and the paragraph immediately following the accuracy theorem in §2.3.

What. The appendix says the two subtracted terms “sum to $1 - \mathbb{E}_{\tilde{\pi}}[w_{\star}]$, *the value quoted after Theorem [accuracy]*”. No value is quoted there — the main text says only that both discrepancies are nonnegative.

Why it matters. The identity is correct. The left side of the surrogate identity evaluates as

$$\int \tilde{\pi} \varrho w_{\star} dx - \int \tilde{\pi} w_{\star} dx = \int \pi dx - \mathbb{E}_{\tilde{\pi}}[w_{\star}] = 1 - \mathbb{E}_{\tilde{\pi}}[w_{\star}],$$

and by the correlation lemma its right side is the sum of the two subtracted discrepancies. So

$$\lambda \mathbb{E}_{x,y \sim \tilde{\pi}}[|w_{\star}(x) - w_{\star}(y)|] + \vartheta \mathbb{E}_{x,y \sim \xi}[|w_{\star}(x) - w_{\star}(y)|] = 1 - \mathbb{E}_{\tilde{\pi}}[w_{\star}] < 1,$$

since $\mathbb{E}_{\tilde{\pi}}[w_{\star}] > 0$. **The improvement the accuracy theorem delivers over $\text{KL}(\pi \parallel \tilde{\pi})$ is strictly less than one nat, whatever λ and ϑ are;** the coefficients redistribute the subtraction between the two weightings rather than enlarging it. This materially qualifies contribution 2, and a referee will find it. Better to state it than to have it found.

Suggested fix. Extend the paragraph after the accuracy theorem (paste-ready):

Both subtracted discrepancies are nonnegative, so the two log-ratio variations improve the error that a fixed biased surrogate leaves behind, under two different weightings of the same ratio $w_{\star}$. Appendix~\ref{sec: appendix-accuracy} evaluates their sum exactly: it

equals $1 - \mathbb{E}_{\tilde{\pi}}[w_{\star}]$. Hence $\mathbb{E}_{\tilde{\pi}}[w_{\star}] \leq 1$, and the improvement is strictly less than one nat at any choice of λ and ϑ ; the coefficients redistribute the subtraction between the two weightings rather than enlarging it.

Or, if you prefer a numbered statement:

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\begin{corollary}
\label{cor: one-nat}
Under the hypotheses of Theorem~\ref{thm: accuracy}, the two subtracted
discrepancies in \eqref{eq: accuracy-bound} sum to exactly
 $1 - \mathbb{E}_{\tilde{\pi}}[w_{\star}]$ . In particular
 $\mathbb{E}_{\tilde{\pi}}[w_{\star}] \leq 1$ , and the improvement over
 $\text{KL}(\pi \parallel \tilde{\pi})$  is strictly less than one nat for every
 $(\lambda, \vartheta)$ .
\end{corollary}
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1.2 Asymptotic unbiasedness is stated without its hypothesis

Where. Abstract (“prove it asymptotically unbiased in the sample size”); contribution 4 of §1.1 (“so the scheme is asymptotically unbiased”); §4.4.

What. The propagation theorem requires the ν_k -essential supremum $\|w_k\|_{\infty}$ to be finite at every stage. The abstract and contribution 4 assert the conclusion unconditionally.

Why it matters. The factor appendix states, correctly, that for a singular potential this supremum “need not be finite at all” — and that is exactly the molecular case §6 is devoted to. If it is infinite at any stage then $F_{\Sigma+} = \infty$ and the propagation bound is vacuous. §6.5 already carries the caveat (“whenever the stage weight suprema are finite”); the front matter should carry the same one.

Suggested fix. Abstract: “We bound that generator’s sampling error and, when the stage weights are essentially bounded, prove it asymptotically unbiased in the sample size.” Contribution 4: append “... provided each stage weight is ν_k -essentially bounded” to the sentence containing the theorem. One clause in each place.

1.3 The sufficient condition of the domination remark excludes the paper’s own defaults

Where. The domination remark; also relevant to the positivity assumption behind the FR flow.

What. The remark offers $\xi \leq c\tilde{\pi}$ together with $\lambda + c\vartheta < \frac{1}{2}$ as a sufficient condition for the two finiteness hypotheses, and notes that $c \geq 1$ necessarily. The paper’s default is $(\lambda, \vartheta) = (1, 1)$, so $\lambda + c\vartheta \geq 2$: the condition can never hold in the configuration every experiment uses.

Why it matters. As written, the only concrete verification offered for the accuracy theorem’s hypotheses is one the paper never satisfies. A reader will also worry about positivity independently: in the FR flow the coefficient of π is $1 + 2\lambda S^{\pi}$, negative wherever $S^{\pi} = -1$ and $\lambda > \frac{1}{2}$, which suggests ν_t can be driven through zero at the default $\lambda = 1$. *It cannot*, and the one-line reason is worth printing.

Suggested fix. Add to the domination remark, and a compressed version after the FR flow (paste-ready):

The condition is sufficient but conservative, and positivity does not in fact require it. If $\nu_\star(x)$ approaches zero at some x with $\pi(x) > 0$, then $w_\star(x) \rightarrow \infty$, so $w_\star(x)$ exceeds $w_\star(y)$ for almost every y and both $S^{\{\tilde{\pi}\}}_\star(x)$ and $S^{\{\xi\}}_\star(x)$ tend to $+1$; the right-hand side of $\text{\eqref{eq: surrogate-stationary}}$ then tends to $\pi(x)(1+2\lambda)+2\vartheta\xi(x) > 0$. Positivity is therefore self-enforcing at any $\lambda, \vartheta \geq 0$. The same argument applies to the flow $\text{\eqref{eq: FR-flow}}$: where ν_t is small the importance weight is large, both correlations approach $+1$, and the drift approaches $\pi(x)(1+2\lambda)+2\vartheta\xi(x) > 0$.

1.4 The exponential rate is forward KL’s own, and the abstract does not say so

Where. Abstract (“We derive exponential convergence in Fisher–Rao geometry”); §2.2; conclusions.

What. In the dissipation identity the two variation terms are nonpositive and are then *discarded*; the decay $\text{KL}(\pi\|\nu_t) \leq e^{-t}\text{KL}(\pi\|\nu_0)$ follows from $\chi^2 \geq \text{KL}$ alone and holds verbatim at $\lambda = \vartheta = 0$.

Why it matters. The body says this plainly — “the rate is the one forward KL already carries” — and the honesty is a strength. But the abstract, read alone, presents exponential convergence as something established *for KLXX*, inviting the objection that the analysis does not discriminate between the losses. The discriminating result is the accuracy theorem, and the abstract should point there.

Suggested fix. Abstract: “We derive convergence in Fisher–Rao geometry at forward KL’s own exponential rate, uniformly in the loss coefficients, together with a fixed-surrogate error bound that the log-ratio variations strictly improve.” Optionally close §2.2 with: “The variations enter the dissipation only as additional nonpositive terms; what distinguishes them is §2.3, not this rate.”

1.5 The finite-sample explanation is applied to one chiral target and not the other

Where. §6.5, the 2,3-butanediol paragraph and the “Ac-Pro-NHMe residual gap” paragraph.

What. The Ac-Pro-NHMe cis-population gap of 0.0244 is attributed to the finite sample. The 2,3-butanediol trans-population gap of 0.0197 is noted as exceeding the reference’s own sampling error (≈ 0.003) and then left without an explanation.

Why it matters. The heuristic error scale $\hat{F}_\Sigma/(2\sqrt{N})$, read from the propagation bound at $\text{osc}(\varphi) = 1$, happens to support the Ac-Pro reading and to strain the butanediol one:

target	$\hat{F}_\Sigma$	N	$\hat{F}_\Sigma/(2\sqrt{N})$	observed gap
alanine dipeptide	68.08	10^7	0.0108	0.0018
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol	37.45	10^7	0.0059	0.0197
Ac-Pro-NHMe	176.7	1.5×10^7	0.0228	0.0244

Applying the estimate only where it helps is the kind of asymmetry a referee notices. Since $\hat{F}_\Sigma$ is explicitly not a bound, neither row proves anything — but the two paragraphs should be written to the same standard.

Suggested fix. Either (i) drop the quantitative finite-sample framing from the Ac-Pro paragraph and say only that the theorem makes the gap an $N^{-1/2}$ term whose size the factor does not predict; or (ii) give the scale for all three targets in one sentence and say outright that the butanediol residual sits above it, so its origin is open. Option (ii) is stronger: the paper already leaves the Ac-Pro gap open rather than closing it by tuning, and the same candour costs nothing here.

1.6 Notation: q carries two meanings two equations apart

Where. §6.1: q_i, q_j are atomic partial charges in the Amber energy; q is the internal-coordinate vector in $x = x(q)$ in the reduced potential and everywhere after.

What. Two distinct objects, same letter, adjacent equations. A third q_k appears in the proof of the one-step lemma, though that one is local to a two-line argument.

Suggested fix. Rename the partial charges to Q_i in the Amber energy and in the sentence listing the nonbonded parameters. The Markov kernel Q_k lives only in the propagation appendix and never meets §6, so no new collision is created. If you prefer to avoid Q altogether, q_i or c_i also work. Separately, rename q_k in the one-step lemma’s proof to δ_k .

1.7 The propagation theorem is attached to a scheme no numbered object states

Where. The propagation theorem and §4.4; §6.5 and the conclusions, both of which say “the inference of Algorithm [staging]”.

What. The staging algorithm is the *training* procedure. The inference pass — push, weight, resample, rejuvenate, with frozen maps on a fresh particle set — is the unnumbered paragraph after it, and that is what the theorem is about.

Why it matters. Two readings are then possible, and the wrong one is the more natural: that the theorem covers the run in which the schedule and the flow-versus-identity choice were made using the same particles they then propagate. It does not, and the propagation appendix says so (“no further selection or training”). The main text does not.

Suggested fix. Number the inference pass — as a further algorithm, or as a labelled display — and point the theorem, §6.5 and the conclusions at it. Then add one sentence to §4.4 (paste-ready):

The theorem concerns inference with the trained maps held fixed, on a fresh particle set. It does not cover the stage selection and the flow-versus-identity comparison of Algorithm~\ref{alg: boltzmann}, which are made using the same particles they then propagate. The inference sets reported in Section~\ref{sec: molecular} are produced by replaying the frozen stage flows on independent particles, which is the setting the theorem describes.

1.8 The stated default ESS gate is never the one used

Where. §4.3 step (iv) (“with default $\tau_v = 0.6$ ”) and the input line of the staging algorithm.

What. Every reported experiment uses $\tau_v = 0.4$. The clock model flags this as nondefault; the molecular runs say “the ESS gate is 0.4 throughout” without flagging it.

Suggested fix. Either change the stated default to 0.4, or add “the experiments of §5–§6 use $\tau_v = 0.4$ ” where the default is introduced. The same applies, less sharply, to $t_{\text{safe}} = 0.2$, which no reported run uses (0.25 or 0.1 instead).

2 Minor

1. **Two ratios round the other way.** §6.4 reports wall-time ratios 2.02, 2.16, 1.86. From the achiral table: $12.1/6.0 = 2.02 \checkmark$, but $35.8/16.5 = 2.170$ and $43.1/23.1 = 1.866$, which round to 2.17 and 1.87. Presumably computed from unrounded times — quote the times to one more digit, or adjust the ratios.
2. **Leftover commented block.** After §4 there is a commented-out `\bibliographystyle / \bibliography / \end{document}` group, evidently from splitting the draft. Harmless, but delete it before submission.
3. **“Stage time” versus “wall time”.** The chiral table says “total stage time”; the alkane and achiral tables say “total wall time”. §1.1 then says the alanine dipeptide run “completes all ten accepted training stages in 72.7 minutes”. If stage time excludes setup or inference, the introduction overstates. Make the three captions agree and say which quantity the introduction quotes.
4. **“Zero” should be “constant”.** §1: “the log-ratio $z(y)$ should be almost surely *zero* on the support of ω ”. The variation penalizes only the failure of z to be *constant* on $\text{supp } \omega$; $z \equiv 0$ follows only when that support is all of $\mathbb{R}^d$. The next clause — “the difference allows the cancellation of the unknown normalizers” — is about constancy, so the two halves of the sentence disagree.
5. **“Decays as” is a bound.** Conclusions: “a gradient flow along which the forward KL decays as e^{-t} ”. The exponential decay is $\text{KL}(\pi \| \nu_t) \leq e^{-t} \text{KL}(\pi \| \nu_0)$, an upper bound, not an asymptotic rate. Write “decays at least as fast as e^{-t} ”.
6. **Tie rule missing from the algorithm.** §6.2 specifies “ties retaining the identity”; the staging algorithm says only “Keep the larger-ESS pair”. Put the tie rule in the algorithm, where an implementer will look for it.
7. **Termination is not argued.** Accepted increments may shrink geometrically without bound, so nothing in the statement of the staging algorithm forces $t_k \rightarrow 1$ in finitely many stages. One sentence closes the acceptance half: as $\Delta t_k \rightarrow 0$ the identity proposal’s weights tend to one, so its ESS tends to 1 and any gate $\tau_v < 1$ is eventually cleared in exact arithmetic. That still does not bound K , which is worth saying rather than leaving implicit.
8. **Cross-reference.** §6.5 says “the reference simulations of Section [alkane results] advance their trajectories sequentially in time”, while discussing chiral references defined in §6.5 itself. Either generalize the sentence or point it at the right subsection.
9. **E_{vac} is defined and then dropped.** The energy decomposition lists the components of E ; the Amber energy defines E_{vac} as the bonded plus vacuum-nonbonded part. The relation $E = E_{\text{vac}} + E_{\text{GB}}^{\text{OBC1}} + E_{\text{ACE}}$ is implied but never written, and E_{vac} never appears again.

10. **Other symbol reuse**, all contextually resolvable but each costing the reader a beat: e is both the energy threshold in $\rho = (e, r)$ and the base of e^{-t} , with the melt scale m_e nearby; r is the pair floor, the radius in the NND_κ definition, and the evaluation point in the first-variation computation of the variation lemma; γ is the shrink factor while γ_τ is a torsion phase; α, β are the mixture coefficients, the α -divergence order in §1.2, and the $\alpha_L/\beta/\alpha_R$ Ramachandran basins in §6.5; B is the batch size while B_t is the Brownian motion in the temper dynamics; and p is the clock state count, the resampling probability, the minority weight p_+ , the LDR exponent, the sector fraction p_s , and the set probability in the two-point example of the factor appendix. A symbol table would earn its half page.

A Checks performed

Derivations re-derived and confirmed

Sign conventions of the pushforward density, the reverse and forward KL, and the log-ratio; the first variation $\delta\mathcal{X}_\omega/\delta\nu = -2(\omega/\nu)S^\omega$ and $\text{grad}^{\text{FR}}\mathcal{X}_\omega = -2\omega S^\omega$ (the variation lemma), including the antisymmetry step and the vanishing mean subtraction; the correlation lemma; the dissipation identity with its coefficient collapse $2\lambda \cdot \frac{1}{2} = \lambda$; $\chi^2 \geq \text{KL}$ from $\log u \leq u - 1$; Grönwall; the accuracy theorem in full, including absolute convergence of the surrogate identity and the equality case; the propagation theorem in full — the domination induction, the one-step lemma, the $2\|w_k\|_\infty + \|w_k\|_\infty$ accounting that produces the factor 3, the conditional-i.i.d. claim under multinomial resampling, both applications of Popoviciu’s inequality, and the unrolling.

Also confirmed: the LDR sandwich $\mathbb{E}_\pi|z - \mathbb{E}_\pi z| \leq \mathcal{X}_\pi \leq 2\mathbb{E}_\pi|z - \mathbb{E}_\pi z|$ in both directions (the factor 2 is valid though not attained — a Gaussian gives $\sqrt{2}$); $F_\Sigma \leq F_{\Sigma+}$; the two-point example with $p = \Lambda^{-3}$; the degenerate identity-kernel case $F_{\Sigma+} = (3^{K+1} - 1)/2$ with $C \sim 9^K$; the $\frac{B-1}{B}$ permutation bias of the empirical variation; C^1 continuity of the soft cap at $\Delta E = e$; and the directions of the endpoint-to-physical weight and the sharpening weight.

Numbers recomputed

quantity	in paper	recomputed
product-well barrier $12 - \frac{1}{2} \ln 24 - \frac{1}{2}$	≈ 9.9	9.911
double-well minima $\pm\sqrt{\ln 24}$	as stated	confirmed
ϕ^4 on-site term $\phi^2 + \frac{1}{2}(\phi^2 - 1)^2$	single minimum	$= \frac{1}{2}\phi^4 + \frac{1}{2}$
hL^2 at (6, 0.0257) and (8, 0.0144)	≈ 0.92	0.9252, 0.9216
$d = 3A - 6$, all nine target rows	9–72	all confirmed
$\hat{F}_\Sigma$, clock $B=2000$, KL / KLXX	21.4 / 15.6	21.413 / 15.625
$\hat{F}_\Sigma$, clock $B=250$, KL / KLXX	62.7 / 41.7	62.692 / 41.728
geometric mean stage ESS, $B=2000$	0.556 / 0.665	0.5556 / 0.6650
$\hat{F}_\Sigma$ reduction ratios, clock	1.33–1.50	1.335–1.504
clock stage schedules, all four batch sizes	clock table	reproduce exactly from $\gamma=0.7$, $\Gamma=2$, $t_{\text{safe}}=0.25$
floor $\hat{F}_\Sigma \geq 2K + 1$, every molecular row	—	all satisfied (methane 7.033 vs. 7)
KLXX fixed / sharpening factor ratios	1.03, 1.04, 1.19, 1.66	1.031, 1.036, 1.192, 1.662
alkane wall-time ratios	3.10, 2.24, 2.65	3.094, 2.238, 2.648
achiral $\hat{F}_\Sigma$ reductions	1.34, 2.70, 2.15	1.335, 2.699, 2.151
ADP: sparse region factor below β/PPII	≈ 240	240.6

quantity	in paper	recomputed
ADP: max population / free-energy difference	0.0018 / 0.03	0.0018 / 0.030
butanediol: max rotamer / free-energy difference	0.020 / 0.23	0.0197 / 0.227
butanediol: reference sampling error	“near 0.003”	0.0031
Ac-Pro: ΔG from populations, $k_B T=2.494$	4.97 / 4.44	4.972 / 4.439
Ac-Pro: accepted jumps $2 \times 10^6 \times 0.2028$	4.06×10^5	4.056×10^5
occupancy-bias slopes (endpoint estimate)	$-0.453 / -0.545$	$-0.451 / -0.558$
occupancy test work per point, $2^k 10^4 \cdot 2^{8-k}$	2.56×10^6	confirmed
2D and ϕ^4 table entries vs. in-text values	—	all consistent

Not checkable from the source: figure contents; the 0.644%, 1.058% and 0.051 alkane-versus-reference differences; the $26 k_B T$ amide barrier; and any claim resting on a run’s output rather than on a tabulated number.